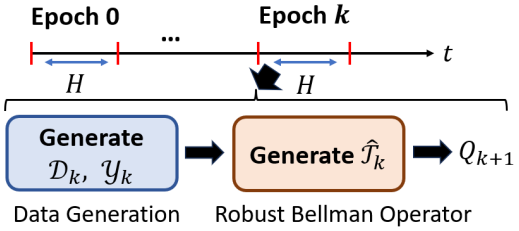
Robust Asynchronous Q-Learning under Reward and State Corruption via Batching

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Problem and method. A learner receives one transition at a time while rewards and next-state reports may be corrupted. **BR-Async-Q** freezes Q_k for H observations, robustly estimates rewards and look-ahead values separately, and makes one Q-table update per epoch. Batching removes visitation-probability amplification from the corruption bias.



Robust Bellman estimation with a frozen Q-table.

Assumptions and notation. Current state-action labels are clean and sampled i.i.d. from $\lambda(s,a) > 0$; set $\lambda_{\min} := \min_{s,a} \lambda(s,a)$. Rewards satisfy $|R(s,a)| \le \bar{R}$ and $\mathrm{Var}(r|s,a) \le \bar{\sigma}^2$; let $\widetilde{\sigma} := \max\{\bar{R}, \bar{\sigma}\}$. Independent-in-time Huber indicators corrupt rewards with probability ϵ_R and next-state reports with probability ϵ_Y; injected values may depend on the past.

Epoch update. For each pair, robustify rewards and bounded look-ahead values separately:

$$\widehat{R}_k = \mathrm{TRIM}_H(D_k, \epsilon_R, \delta_1), \qquad \widehat{\mu}_k = \mathrm{TRIM}_B(Y_k, [-B, B]), \qquad Q_{k+1} = (1-\alpha)Q_k + \alpha\big[\mathrm{clip}_{[-G_k, G_k]}(\widehat{R}_k) + \gamma\widehat{\mu}_k\big].$$

Here $\delta_1 = \delta/(4SAT)$, Y_k contains $\max_{a'} Q_k(\widetilde{s}', a')$, $B = 3C\widetilde{\sigma}/(1-\gamma)$, and G_k controls rare estimation failures.

1. High-probability finite-time guarantee. With admissible trimming levels, choose

$$K = \left\lceil \frac{2\log T}{1-\gamma} \right\rceil, \qquad \alpha = \frac{\log T}{(1-\gamma)K}, \qquad T = KH \ge K\left\lceil \frac{64}{\lambda_{\min}} \log \frac{8TSA}{\delta} \right\rceil.$$

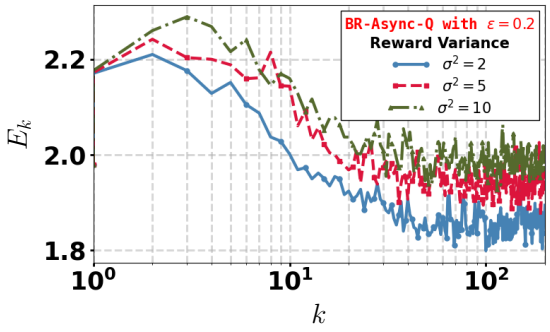
Writing $e_k := \|Q_k - Q^\star\|_\infty$, with probability at least $1-\delta$,

$$e_K \le \frac{e_0}{T} + \mathcal{O}\left(\frac{\widetilde{\sigma}\sqrt{\log T \log(TSA/\delta)}}{(1-\gamma)^{5/2}\sqrt{\lambda_{\min} T}} \right) + \mathcal{O}\left(\frac{\bar{\sigma}\sqrt{\epsilon_R}}{1-\gamma} \vee \frac{\widetilde{\sigma}\epsilon_Y}{(1-\gamma)^2} \right). \tag{1}$$

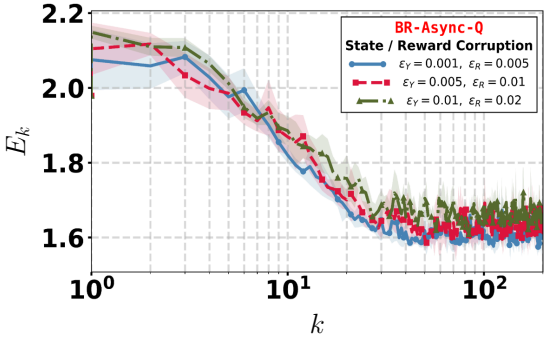
2. Optimal reward-corruption dependence. For $\epsilon_Y = 0$, the bias is $\mathcal{O}(\bar{\sigma}\sqrt{\epsilon_R}/(1-\gamma))$, matching the information-theoretic reward-corruption lower bound. Neither corruption term depends on attack magnitude or on $\lambda_{\min}$.

3. Improved statistical dependence on visitation. The clean term scales as $1/\sqrt{\lambda_{\min} T}$. Long epochs supply enough visits to every pair for robust estimation before each frozen-table update.

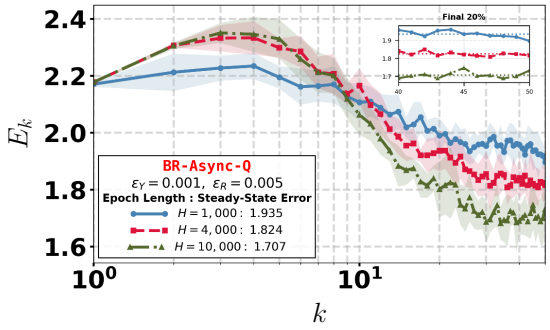
Numerical results. Synthetic grid-world: $S = 100$, $A = 5$, $\gamma = 0.5$, $\alpha = 0.1$; $E_k = \|Q_k - Q^\star\|_\infty$. Panels (c,d) vary $H \in \{1000, 4000, 10000\}$.



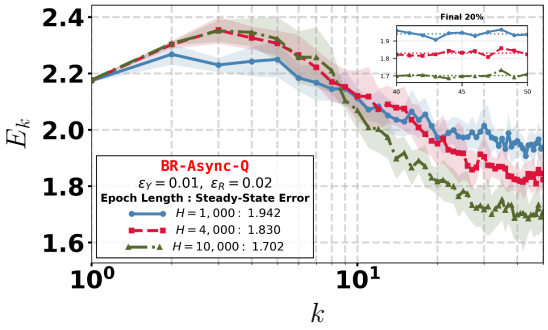
(a) Reward variance; $\epsilon = 0.2$.



(b) Joint reward and next-state corruption.



(c) Epoch length; $(\epsilon_Y, \epsilon_R) = (0.001, 0.005)$.



(d) Epoch length; $(\epsilon_Y, \epsilon_R) = (0.01, 0.02)$.